

11.1 Why Colour Management?

Ever printed out a holiday photo only to find that the print copy hardly looks as good as the one on your screen? Countless expletives have been uttered in such situations, all in effect asking the same question: Why do images look so different on my monitor and when they finally print?

If you've used your old film camera often, you might have noticed that the brand of film you choose can have a distinct effect on the way your prints appear—some brands will give you rich, saturated colours, some will cause skin tones to look darker than they are, and so on: different chemicals are affected differently by the same colours. Similarly, different devices will interpret the same real-life colour in the way they understand it.

Colour Spaces and The ICC

A device's *Colour Space* is the entire range (or gamut) of colours that the device is capable of reproducing. You can deduce that these colour spaces are highly dependant on the device and its eccentricities—even two otherwise identical printers might have different colour spaces!

Then there's the editing colour space, which defines the colour range you do your work in. Editing colour spaces don't depend on devices at all, so there's no restriction on which colours you can work with. And this is where things start to go wrong—the moment you decide to use a colour that isn't in your printer's colour space, you'll end up with a sub-standard image.

Thankfully, the International Color Consortium (ICC) has come to our rescue and set a standard profile (called an ICC profile) that defines how devices treat colour. Printer profiles will tell Photoshop (or any other image editing program) what the printer is capable of, and will let Photoshop decide how to accurately bring the image on your screen to paper. The editing profiles provided by the ICC tell you which colours it is safest to